CHAITANYA PRABHU

SYSTEMS ENGINEER

New Delhi, India

Systems engineer working close to the metal — distributed systems, network programming, and the operating-system primitives beneath them. I build consensus, replication, and servers from raw POSIX sockets up in C and C++: explicit memory and error paths, careful lock discipline, and durability you can defend. I care about correctness under failure and prove it with chaos testing and sanitizers rather than hope.

EDUCATION

IIIT Hyderabad — M.Tech, Computer Science 2025 – PRESENT

GPA 8.2 / 10 · Coursework: Advanced Operating Systems, Software System Development, Data Structures & Algorithms, System & Network Security.

NIT Silchar — B.Tech, Civil Engineering 2020 – 2024

GPA 7.4 / 10

TECHNICAL SKILLS

LANGUAGES	C, C++17, Python, TypeScript / JavaScript, SQL, Bash
SYSTEMS	POSIX sockets, pthreads, fork / exec, signals, timerfd, O_DSYNC, positional I/O, Linux internals
TOOLING	ASan / UBSan / TSan / LeakSanitizer, GDB, Make, Docker, Kubernetes, AWS, Redis, PostgreSQL
FOCUS	Consensus & replication, network protocols, performance engineering, applied cryptography

PROJECTS

SYSTEMS & NETWORKING

Peer-to-Peer File-Sharing Distributed System C++17 · 2026

- BitTorrent-style content distribution on raw POSIX sockets with zero dependencies; a control / data-plane split keeps a 2-node tracker cluster holding metadata only while 512 KiB pieces move peer-to-peer, with idempotent op-based replication that self-heals on reconnect and transparent fail-over via RPC retry and session replay.
- Rarest-first scheduling over a multi-connection pool with per-piece SHA-1 verify-before-commit; O(1)-in-file-size memory via positional I/O and streaming hashes — a 200 MB transfer in ~7 MB RSS. Verified clean under ASan / UBSan / LeakSanitizer.

C++17 · POSIX SOCKETS · OP-BASED REPLICATION · SHA-1

POSIX Shell C++17 · 2026

- Interactive Unix shell built directly on process-management primitives: core built-ins, I/O redirection, arbitrary-depth pipelines, background job control, full signal handling, tab completion, and persistent navigable history.

C++17 · FORK / EXEC · PIPES · SIGNALS

SECURITY & CRYPTOGRAPHY

Secure UAV Command & Control System C++17 · 2026

- Encrypted command-and-control between a Mission Control Center and a drone fleet; hand-implemented 2048-bit ElGamal, HMAC-SHA256, and AES-256-CBC for mutual authentication, session- and group-key establishment, and fleet-wide broadcast — sub-500 ms handshake after key generation.

C++17 · ELGAMAL 2048-BIT · HMAC-SHA256 · AES-256-CBC

Kerberos Under Partial Compromise PYTHON · 2026

- Kerberos-inspired authentication resilient to single-authority compromise via a 2-of-3 Schnorr multi-signature across distributed Authentication / Ticket-Granting servers; AES-256-CBC tickets; ships attack simulations for key leakage and offline-authority compromise.

PYTHON · SCHNORR MULTI-SIGNATURES · AES-256-CBC · THRESHOLD CRYPTOGRAPHY

Secure Multi-Client Communication PYTHON · 2026

- Stateful, symmetric-key secure-communication protocol between a server and multiple clients, designed to hold up in a hostile network environment.

PYTHON · SYMMETRIC-KEY CRYPTOGRAPHY · SESSION STATE

EXPERIENCE

Engineer Intern — Deendayal Port Authority MAY – NOV 2023

Gandhidham, Gujarat · On-site

- Maintained the relational cargo-logging database for Oil Jetty No. 8 on the port's Linux servers — wrote SQL and PL/SQL stored procedures and triggers to record, reconcile, and audit consignment and vessel data, enforcing validation and referential integrity across high-volume daily entries.

ACHIEVEMENTS

- **GATE 2025** CS & IT — 59.62 / 100 · **GATE 2024** DS & AI — 62.67 / 100 · **JEE Main 2020** — 98.28 percentile
- Winner, All-India Inter-NIT Chess (2022) · Runner-up, All-India Inter-NIT Football (2023)
- Head, Publication Team — Technoesis 2023, NIT Silchar